H.6 Inflation Indices

Chapter 4 of the BenMAP-CE User Manual provides instructions for formatting and adding inflation data. These values are used to adjust economic values to express monetary units in a consistent dollar year. As discussed in that chapter, BenMAP-CE includes inflation factors for three different types of values. The source for these values is included in Table H-17. These values were re-indexed to $2015 prior to import in BenMAP-CE.

Table H-17. Inflation Factors

Name	Description	Years	Source
All Goods Index	Value of generic goods and services	1980-2022	BLS, Data Series CUUR0000SA0 at http://data.bls.gov/cgi-bin/surveymost?cu
Medical Cost Index	Value of medical expenses		BLS, Data Series CUUR0000SAM at http://data.bls.gov/cgi-bin/surveymost?cu
Wage Index	Value of wages		BLS, Employment Cost Trends. Table 5 at http://www.bls.gov/web/eci/ecicois.txt

Appendix I. Additional Health Valuation Functions in U.S. Setup

In this Appendix, we present additional health valuation functions. Unlike those in Appendix H, these functions are included in the U.S. Setup but are not currently used by the U.S. EPA in regulatory impact analyses. For the health valuation functions currently used by EPA, see the following page: https://www.epa.gov/benmap/benmap-community-edition. For Ozone Health Valuation Functions, click the "U.S. EPA approach for quantifying and valuing ozone effects" link. For $PM_{2.5}$ Health Valuation Functions, click the "U.S. EPA approach for quantifying and valuing PM effects" link.

I.1 Mortality

I.1.1 Value of a Statistical Life Based on Selected Studies

In addition to the value of a statistical life based on the results of 26 studies in Appendix H, section H.1.1, we have included three alternatives based loosely on the results of work by Mrozek and Taylor (2002) and Viscusi and Aldy (2003). Each of these three alternatives has a mean value of $7.6 million (2015$), but with a different distribution: normal, uniform, triangular, and beta. Table I-1 presents the distribution parameters for these additional valuations in BenMAP.

Table I-1. Additional Unit Values for VSL for additional value-of-life studies

Basis for Estimate *	Age Range at Death		Unit Value (VSL) (2015$)	Distribution of Unit Value	Parameters of Distribution	
	Min	Max			P1	P2
VSL, based on 2015$ range from $1.38 million to $13.76 million – 95% CI of assumed normal distribution	0	99	7,570,229	Normal	3,160,172	–
VSL based on 2015$ range from $1.38 million to $13.76 million – assumed uniform distribution	0	99	7,570,229	Uniform	1,376,405	13,764,053
VSL based on 2015$ range from $1.38 million to $13.76 million – assumed triangular distribution	0	99	7,570,229	Triangular	1,376,405	13,764,053

* The original value of a statistical life was calculated in 1990 $. We have used a factor of 1.8134, based on the All-Items CPI-U.

I.2 Hospital Admissions

This sub-section presents the unit values for hospital admissions that are not used by the EPA in regulatory impact analyses but are included in BenMAP. See Appendix H, Section H.2 for more information about these unit values. Table I-2 includes the unit values for hospital admissions for endpoints included in BenMAP but not used by the EPA in their regulatory impact analyses.

Table I-2. Additional Unit Values Available for Hospital Admissions

Endpoint	ICD Codes	Age Range		Mean Hospital Charge (2015 $)	Mean Length of Stay (days)	Total Cost of Illness (Unit Value in 2015$)*
		Min	Max			
HA, All Cardiovascular	390-429	0	99	$33,063	4.59	$33,856
HA, All Cardiovascular	390-429	0	64	$45,659	4.12	$46,371
HA, All Cardiovascular	390-429	65	99	42,642	4.88	43,485
HA, Asthma	493	0	64	$16,655	3.00	$17,174
HA, Chronic Lung Disease	490-496	18	64	$21,989	3.90	$22,663
HA, Congestive Heart Failure	428	65	99	$33,734	5.32	$34,654
HA, Dysrhythmia	427	0	99	$33,063	3.72	$33,706
HA, Ischemic Heart Disease	410-414	65	99	$55,591	4.61	$56,388
HA, All Respiratory	460-519	0	1	$16,929	3.19	$17,480
HA, All Respiratory	460-519	0	99	$32,563	5.35	$33,488
HA, Asthma	493	65	99	$26,153	4.79	$26,981
HA, Asthma	493	0	99	$18,590	3.37	$19,172
HA, Chronic Lung Disease	490-496	65	99	$25,413	4.79	$26,241
HA, Chronic Lung Disease	490-496	0	99	$22,312	4.10	$23,021
HA, Chronic Lung Disease (less Asthma)	490-492, 494-496	18	64	$23,980	4.23	$24,711
HA, Chronic Lung Disease (less Asthma)	490-492, 494-496	65	99	$25,254	4.79	$26,082
HA, Chronic Lung Disease (Less Asthma)	490-492, 494-496	0	99	$24,834	4.59	$25,627
HA, Pneumonia	480-487	65	99	$30,229	5.77	$31,226
HA, Pneumonia	480-487	0	99	$29,046	5.25	$29,953

* The opportunity cost of a day spent in the hospital was estimated, for the above exhibit, at the median daily wage of all workers, regardless of age. The median daily wage was calculated by dividing the median weekly wage ($864 in 2015$) by 5. The median weekly wages for 2015 were obtained from the U.S. Census Bureau's 2015 American Community Survey, "Selected Economic Characteristics: 2015 American Community Survey 1-Year Estimates."

I.3 Chronic Illness

This sub-section presents the unit values developed for chronic bronchitis, chronic asthma, and non-fatal myocardial infarctions.

I.3.1 Chronic Bronchitis

PM-related chronic bronchitis is expected to last from the initial onset of the illness throughout the rest of the individual's life. WTP to avoid chronic bronchitis would therefore be expected to incorporate the present discounted value of a potentially long stream of costs (e.g., medical expenditures and lost earnings) as well as WTP to avoid the pain and suffering associated with the illness. Both WTP and COI estimates are currently available in BenMAP.

I.3.1.1 Unit Value Based on Two Studies of WTP

Two contingent valuation studies, Viscusi et al. (1991) and Krupnick and Cropper (1992), provide estimates of WTP to avoid a case of chronic bronchitis. Viscusi et al. (1991) and Krupnick and Cropper (1992) were experimental studies intended to examine new methodologies for eliciting values for morbidity endpoints. Although these studies were not specifically designed for policy analysis, they can be used to provide reasonable estimates of WTP to avoid a case of chronic bronchitis. As with other contingent valuation studies, the reliability of the WTP estimates depends on the methods used to obtain the WTP values. The Viscusi et al. and the Krupnick and Cropper studies are broadly consistent with current contingent valuation practices, although specific attributes of the studies may not be.

The study by Viscusi et al. (1991) uses a sample that is larger and more representative of the general population than the study by Krupnick and Cropper (1992), which selects people who have a relative with the disease. However, the chronic bronchitis described to study subjects in the Viscusi study is severe, whereas a pollution-related case may be less severe.

The relationship between the severity of a case of chronic bronchitis and WTP to avoid it was estimated by Krupnick and Cropper (1992). We used that estimated relationship to derive a relationship between WTP to avoid a severe case of chronic bronchitis, as described in the Viscusi study, and WTP to avoid a less severe case. The estimated relationship (see Table 4 in Krupnick and Cropper) can be written as:

$$1n(WTP) = \alpha + \beta \times sev$$

where a denotes all the other variables in the regression model and their coefficients, ß is the coefficient of sev, estimated to be 0.18, and sev denotes the severity level (a number from 1 to 13). Let x (< 13) denote the severity level of a pollution-related case of chronic bronchitis, and 13 denote the highest severity level (as described in Viscusi et al., 1991). Then

$$1n(WTP_{13}) = \alpha + \beta \times 13$$

and

$$1n(WTP_{x}) = \alpha + \beta \times x$$

Subtracting one equation from the other,

$$1n(WTP_{13}) - 1n(WTP_{x}) = \beta \times (13 - x)$$

or